Internal energy and The First Law

Internal energy: Energy not kinetic or potential

1) thermal energy (heat)
2) mechanical/strain energy
3) Chemical energy
4) Magnetization

Note: everything below is spatial = Eulerian

Internal energy $\mathcal{U}$ of Ω_t: $\mathcal{U}[\Omega_t] = \int_{\Omega_t} \rho\, u\, dV_x$

$u(\underline{x},t)$ = internal energy density / unit mass

First Law of Thermodynamics

$$\boxed{\frac{d}{dt}\mathcal{U}[\Omega_t] = Q[\Omega_t] + W[\Omega_t]} \qquad \text{for all } \Omega_t \subseteq B_t$$

Net heating: $Q[\Omega_t]$ = rate of thermal energy gain/loss

Net working: $W[\Omega_t]$ = rate of mechanical energy gain/loss

Unit of heating/working/power: Watt $= W = \frac{J}{s}$

Net heating

2 forms of heat gain / loss:

I) Body heating : $Q_b[\Omega_t] = \int_{\Omega_t} \rho H \, dV$

H = heat gain/power per unit mass $\left[\frac{W}{kg}\right]$ $W = \frac{J}{s}$

Example: radiogenic heating

$$H = H_0 e^{-\lambda t}$$

H_0 = initial power per unit mass $\left[\frac{W}{kg}\right]$

λ = decay constant $\left[\frac{1}{s}\right]$

Note: $A = \rho H$ power per unit volume $\left[\frac{W}{m^3}\right]$

II) Surface heating: $Q_s[\Omega_t] = -\int_{\partial\Omega_t} \underline{q} \cdot \hat{\underline{n}} \, dA$

$\underline{q}(\underline{x},t)$ = heat flux vector $\left[\frac{W}{m^2}\right]$

Fourier's law : $\underline{q} = -\kappa \nabla T$ (Lecture 10)

κ = thermal conductivity $\left[\frac{W}{m\,K} = \frac{J}{m\,s\,K}\right]$

<u>Net heating</u> is the rate at which heat is added or lost from Ω_t

$$\boxed{Q[\Omega_t] = Q_b[\Omega_t] + Q_s[\Omega_t] = \int_{\Omega_t} \rho H \, dV - \int_{\partial\Omega_t} \underline{q} \cdot \hat{\underline{n}} \, dA}$$

Energy balance with net heating

First law with $W=0$: $\quad \frac{d}{dt}\mathcal{U}[\Omega_t] = Q[\Omega_t]$

where $\mathcal{U}[\Omega_t] = \int_{\Omega_t} \rho u \, dV_x$

$$Q[\Omega_t] = \int_{\Omega_t} \rho H \, dV_x - \int_{\partial\Omega_t} \underline{q}\cdot\underline{n} \, dA_x$$

substituting:

$$\frac{d}{dt}\int_{\Omega_t} \rho u \, dV_x = -\int_{\partial\Omega_t} \underline{q}\cdot\underline{n}\, dA_x + \int_{\Omega_t} \rho H \, dV_x$$

using derivative relative to mass and divergence Thm

$$\int_{\Omega} (\rho \dot{u} + \nabla_x\cdot \underline{q} + \rho H) \, dV_x$$

by the arbitraryness of Ω_t we have

$$\boxed{\rho\dot{u} + \nabla\cdot\underline{q} = \rho H} \qquad \text{local Eulerian form}$$

Following lin. mom. balance $\longrightarrow$ conservative form

$$\boxed{\frac{\partial}{\partial t}(\rho u) + \nabla\cdot[\underline{v}\rho u + \underline{q}] = \rho H} \qquad \Rightarrow \text{HW 9}$$

- still in terms of internal energy density!

Relate u to measurable quantities

Fundamental thermodynamic relation: $U = U(S, V)$

$\boxed{dU = T\,dS - p\,dV}$ thermo book version

U = internal energy [J]

S = entropy $\left[\frac{J}{K}\right]$ V = volume $[m^3]$

T = temperature [K] p = pressure [Pa]

$\Rightarrow$ good for global statements (whole system)

Continuum mechanics is local

$u = \frac{U}{m}$ = specific internal energy $\left[\frac{J}{kg}\right]$

energy/mass m = mass

need $u = u(T, p)$

$\boxed{du = c_p\,dT - \frac{\alpha T}{\rho}\,dp}$ → take a good thermo class.

(Legendre transformations
& Maxwell relations)

c_p = specific heat capacity $\left[\frac{J}{kg\,K}\right]$

ρ = mass density $\left[\frac{kg}{m^3}\right]$

$\alpha = -\frac{1}{\rho}\left.\frac{\partial\rho}{\partial T}\right|_p$ = isobaric thermal expansivity $\left[\frac{1}{K}\right]$

Advection - Conduction Equation

Energy balance: $\frac{\partial}{\partial t}(\rho u) + \nabla \cdot [\underline{v} \rho u + \underline{q}] = \rho r$

$\underline{v} \rightarrow$ momentum balance

ρ, r parameters (given)

$u, \underline{q}$ unknown variables

1 equation with 2 unknowns

$\Rightarrow$ need to relate u and $\underline{q}$

Assume: $p = \text{const.}$ and $c_p = \text{const.}$

Thermodynamics: $du = c_p \, dT \quad (dp = 0)$

$$u = u_0 + c_p (T - T_0) \qquad c_p = \text{const.}$$

Fouriers law: $\quad \underline{q} = -\kappa \nabla T \qquad$ (Lecture 10)

physical parameters:

1) c_p = specific heat capacity $\left[\frac{J}{kg\,K}\right]$

2) κ = thermal conductivity $\left[\frac{W}{m\,K}\right]$

Assume: 1) properties are constant (ρ, c_p, κ)

2) flow is incompressible $\nabla \cdot \underline{v} = 0$

substitute and simplify

$$\boxed{\rho c_p \frac{\partial T}{\partial t} - \nabla \cdot [\underline{v} \rho c_p T - \kappa \nabla T] = \rho H}$$ $\Rightarrow$ HW9

Advective heat flux: $q_A = \underline{v} \rho c_p T$

Conductive heat flux: $q_c = -\kappa \nabla T$

Heat production: ρH

$\Rightarrow$ Advection - Diffusion - Reaction Equation

Diffusion ~ Conduction

Reaction ~ Production

In limit of $\underline{v} \to \underline{0}$

$$\boxed{\frac{\partial T}{\partial t} - D \nabla^2 T = \frac{H}{c_p}}$$ Heat Equation

where

$\nabla^2 = \nabla \cdot \nabla$ = Laplacian (Lecture 10)

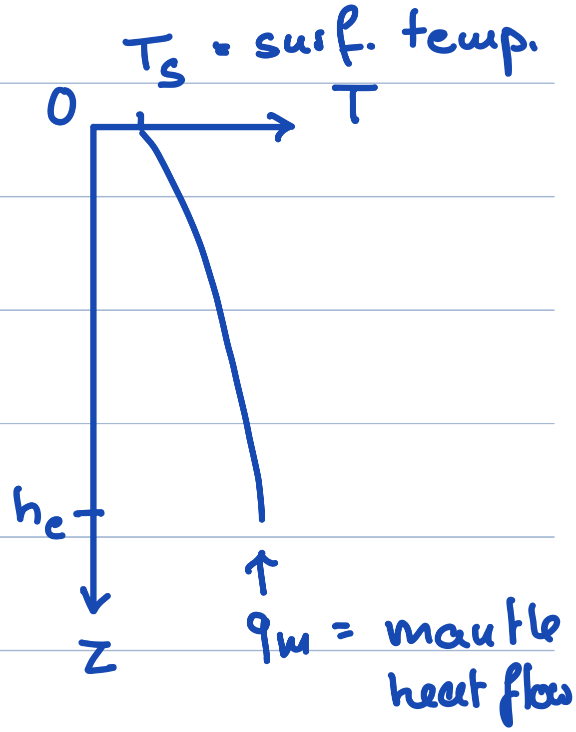
$D = \frac{\kappa}{\rho c_p}$ thermal diffusivity $\left[\frac{m^2}{s}\right]$

$\kappa \to$ heat flux $\qquad$ $D \to$ change in T

Example: Steady State Geotherm

Steady state: $\frac{\partial T}{\partial t} = 0 \Rightarrow -\kappa \nabla^2 T = \rho H$

One dimension: $-\kappa \frac{d^2 T}{dz} = \rho H$



Boundary Conditions:

$T(z=0) =: T_s$

$q(z=H) = -\kappa \left.\frac{dT}{dz}\right|_{z=H} = -q_m$

↑ upward!

Integrate once: $-\kappa \frac{d^2 T}{dz^2} = \rho H \qquad -\kappa \frac{dT}{dz} = \rho H z + c_1$

Use bottom BC: $-\kappa \left.\frac{dT}{dz}\right|_{z=H} = \rho H h_c + c_1 = -q_m$

$\Rightarrow c_1 = -q_m - \rho H h_c$

substitute c_1: $-\kappa \frac{dT}{dz} = \rho H z - \rho r h_c - q_m$

$\Rightarrow$ heat flow: $\boxed{q(z) = -\rho r (h_c - z) - q_m}$

absence of heat flow: $q(z) = -q_m$

Integrate again: $-\kappa T = \rho H \frac{1}{2} z^2 - (\rho H h_c + q_m) z + c_2$

Use top BC: $-\kappa T_s = c_2$

$$\Rightarrow -\kappa T = \frac{\rho H}{2} z^2 - (\rho H h_c + q_m) z - \kappa T_s$$

$\Rightarrow$ geotherm: $\boxed{T(z) = T_s + \left(\frac{\rho H h_c}{\kappa} + \frac{q_m}{\kappa}\right) z - \frac{\rho H}{2\kappa} z^2}$

Chicxulub (Espinosa-Cardena et al. 2016)

$A = \rho r = 1.5 \ \mu W/m^3$ vol. heat production

$\kappa = 2.67 \ \frac{W}{mK}$ thermal conductivity

$h_c = 35$ km

$T_s = 26.5\,°C$ (Merida, Mexico)

$q_m = 28 \ mW/m^2$

$\Rightarrow T(H) = 707$ C $\qquad q(0) = 79 \ mW/m^2$

1) Granite melts at 650°C (amphibolite at 800°C)

$\Rightarrow$ base of crust is close to melting

2) Mean heat flow of continents is 65 mW/m²

$\Rightarrow$ increased heat flow 66 Myrs after impact